

HTML NOTES

HTML & Semantic HTML — Study Reference

Get Techy With Lucky | Instructor: Lucky Nakola

Part 1 — HTML Foundations

What Is HTML?

HTML stands for HyperText Markup Language. It is the standard language used to create the structure of every webpage on the internet.

HTML is NOT a programming language — it is a markup language. It uses special tags to label and describe content, telling the browser what each piece of content is.

ANALOGY: Building a House

HTML = the bricks and concrete — the raw structure

CSS = the paint and decoration — how it looks

JavaScript = electricity and plumbing — the behaviour

You cannot have CSS or JavaScript without HTML first.

Anatomy of an HTML Element

<code><h1></code>	<code>Hello, World!</code>	<code></h1></code>
↑	↑	↑
Opening Tag	Content	Closing Tag

Part	Explanation
Opening Tag	Marks where the element starts. Written as: <code><tagname></code>
Content	The text, image, or nested elements inside.
Closing Tag	Marks where the element ends. Written as: <code></tagname></code>
Attribute	Extra info inside the opening tag: <code></code> . Format: <code>name="value"</code>
Self-closing	Some elements have no content/closing tag: <code></code> , <code>
</code> , <code><hr></code> , <code><input></code>

Standard HTML Document Structure

<pre><!DOCTYPE html> <html lang="en"> <head> <meta charset="UTF-8"> <meta name="viewport" content="width=device-width, initial-scale=1.0"> <title>My Webpage</title> <link rel="stylesheet" href="style.css"> </head></pre>

```
<body>
  <h1>Hello, World!</h1>
  <p>This is my first webpage.</p>
  <script src="app.js"></script>
</body>
</html>
```

Element	What It Does
<!DOCTYPE html>	Declares this as HTML5. Must be the very first line.
<html lang="en">	Root element. lang="en" declares the page language.
<head>	Contains metadata — information about the page, not shown visually.
<meta charset>	Sets character encoding to UTF-8 — supports all languages.
<meta name=viewport>	Controls how the page scales on mobile. Required for responsive design.
<title>	Text shown on the browser tab and in search results (important for SEO).
<link rel=stylesheet>	Connects an external CSS file to this HTML page.
<body>	Contains all visible content the user sees.
<script src>	Links external JavaScript. Placed at END of body so HTML loads first.

Part 2 — Core HTML Elements

Headings — <h1> to <h6>

HTML provides 6 heading levels. They define a content hierarchy like chapters in a book.

```
<h1>Main Page Title — ONE per page only</h1>
<h2>Major Section</h2>
<h3>Sub-section</h3>
<h4>Sub-sub-section</h4>
<h5>Minor detail</h5>
<h6>Smallest heading</h6>
```

HEADING RULES

1. Every page must have exactly ONE <h1>.
2. Never skip heading levels (e.g. h1 → h4).
3. Choose headings for MEANING, not size. Use CSS to control size.
4. Google reads headings to understand your page content (SEO).
5. Screen readers use headings to help blind users navigate.

Paragraphs & Text Formatting

```
<p>A paragraph — block element with space above and below.</p>
<strong>Bold — IMPORTANT text (has semantic meaning)</strong>
<b>Bold — visual only, no semantic meaning</b>
<em>Italic — EMPHASISED text (has semantic meaning)</em>
<i>Italic — visual only, no semantic meaning</i>
<mark>Highlighted text</mark>
<small>Small print — legal text, copyright</small>
<sub>Subscript e.g. H<sub>2</sub>O</sub>
<sup>Superscript e.g. E=mc<sup>2</sup></sup>
<code>Inline code — e.g. console.log()</code>
<br>    <!-- Line break -->
<hr>    <!-- Horizontal rule / thematic break -->
```

Links — <a>

```
<!-- External link -->
<a href="https://google.com">Go to Google</a>

<!-- Open in new tab -->
<a href="https://google.com" target="_blank" rel="noopener noreferrer">New Tab</a>

<!-- Jump to section on same page -->
<a href="#contact">Go to Contact Section</a>

<!-- Email link -->
```

```
<a href="mailto:lucky@example.com">Email Us</a>
```

```
<!-- Phone link -->
```

```
<a href="tel:+254712345678">Call Us</a>
```

Images —

```

```

```
<!-- With dimensions -->
```

```

```

```
<!-- With caption -->
```

```
<figure>
```

```
  
```

```
  <figcaption>The Get Techy With Lucky team — Nyeri, 2025</figcaption>
```

```
</figure>
```

The alt Attribute Is NOT Optional

alt describes what the image shows.

It is read aloud by screen readers for visually impaired users.

It displays if the image fails to load.

It is indexed by Google Images and improves SEO.

Meaningful image: use alt="Lucky teaching HTML in a classroom"

Decorative image: use alt="" (empty string)

Lists

```
<!-- Unordered list — bullet points -->
```

```
<ul>
```

```
  <li>HTML</li>
```

```
  <li>CSS</li>
```

```
  <li>JavaScript</li>
```

```
</ul>
```

```
<!-- Ordered list — numbered -->
```

```
<ol>
```

```
  <li>Open VS Code</li>
```

```
  <li>Create index.html</li>
```

```
  <li>Write your HTML</li>
```

```
</ol>
```

```
<!-- Description list — term/definition pairs -->
```

```
<dl>
```

```
  <dt>HTML</dt>
```

```
  <dd>HyperText Markup Language — the structure of a webpage.</dd>
```

```
</dl>
```

Tables

```
<table>
  <caption>Web Development Languages</caption>
  <thead>
    <tr>
      <th scope="col">Language</th>
      <th scope="col">Role</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>HTML</td>
      <td>Structure</td>
    </tr>
    <tr>
      <td>CSS</td>
      <td>Styling</td>
    </tr>
  </tbody>
  <tfoot>
    <tr><td colspan="2">Both are essential for frontend development.</td></tr>
  </tfoot>
</table>
```

Element	Purpose
<thead>	Table header group — wraps column heading rows.
<tbody>	Table body — wraps all data rows.
<tfoot>	Table footer — wraps totals or summary rows.
<tr>	Table Row — a horizontal row.
<th>	Table Header cell — bold and centred by default. Use scope="col" or scope="row".
<td>	Table Data cell — a regular data cell.
colspan	Makes a cell span multiple columns: <td colspan="3">
rowspan	Makes a cell span multiple rows: <td rowspan="2">

Part 3 — Forms

Forms collect input from users and send it to a server or handle it with JavaScript.

```
<form action="submit.php" method="POST">

  <label for="name">Full Name:</label>
  <input type="text" id="name" name="name" placeholder="Lucky Nakola" required>

  <label for="email">Email:</label>
  <input type="email" id="email" name="email" required>

  <label for="country">Country:</label>
  <select id="country" name="country">
    <option value="ke">Kenya</option>
    <option value="ug">Uganda</option>
  </select>

  <label for="msg">Message:</label>
  <textarea id="msg" name="message" rows="5"></textarea>

  <fieldset>
    <legend>Gender:</legend>
    <input type="radio" id="male" name="gender" value="male">
    <label for="male">Male</label>
    <input type="radio" id="female" name="gender" value="female">
    <label for="female">Female</label>
  </fieldset>

  <input type="checkbox" id="terms" name="terms" required>
  <label for="terms">I agree to the Terms and Conditions</label>

  <button type="submit">Send</button>
  <button type="reset">Clear</button>

</form>
```

Common Input Types

Input Type	What It Does
type="text"	Single-line text entry
type="email"	Email address with format validation
type="password"	Text hidden with dots
type="number"	Numeric input with min/max
type="date"	Date picker calendar

Input Type	What It Does
type="checkbox"	Tick one or many options
type="radio"	Select exactly one from a group
type="file"	Opens file picker dialog
type="range"	Slider between min and max
type="hidden"	Not visible — sends data silently
type="submit"	Button that submits the form

Important Form Attributes

Attribute	Purpose
required	Field cannot be empty when form is submitted.
placeholder	Ghost text shown when the field is empty.
min / max	Min and max values for number, date, range inputs.
minlength / maxlength	Min and max number of characters allowed.
disabled	Field is greyed out and cannot be edited.
readonly	User can see but cannot change the value.
autofocus	Automatically focuses this field on page load.

Part 4 — Semantic HTML

What Is Semantic HTML?

Semantic HTML means choosing elements based on their MEANING — what the content IS — not just how it looks. A semantic element describes its own purpose to the browser, search engines, and assistive technologies.

Approach	What It Means
Non-Semantic	Uses generic <code><div></code> and <code></code> with class names. Browser knows it is a box — nothing more.
Semantic	Uses <code><header></code> , <code><nav></code> , <code><main></code> , <code><article></code> , <code><footer></code> . Browser knows exactly what each part IS.

WHY SEMANTIC HTML MATTERS

- 1. ACCESSIBILITY — Screen readers use semantic elements to help visually impaired users navigate. `<nav>` announces 'this is navigation'. `<main>` lets users skip to content.
- 2. SEO — Search engines use semantic structure to understand your content. `<article>` tells Google 'this is indexable content'. `<h1>` = most important heading.
- 3. MAINTAINABILITY — `<header>` is clearer to read than `<div class="header">`.
- 4. BROWSER BEHAVIOUR — Browsers apply correct default styles automatically.

Page-Level Semantic Elements

`<header>`

Introductory content for a page or section — logo, main navigation, hero area. A page can have multiple `<header>` elements (one per page, article, or section).

`<nav>`

Marks a major navigation block — primary menu, breadcrumbs, pagination. Use `aria-label` when you have more than one `<nav>` to distinguish them.

`<main>`

Contains the primary, unique content of the page. There must be only ONE `<main>` per page. Never nest it inside `<header>`, `<footer>`, `<nav>`, or `<aside>`.

`<section>`

Groups related content sharing a common theme. Each <section> should have its own heading. Use when content belongs together but isn't self-contained enough to be an <article>.

<article>

Self-contained content that could be distributed independently — a blog post, news article, product card, user comment. Ask: 'Does this make sense on its own?' If yes — use <article>.

<aside>

Tangentially related content — a sidebar, pull quote, related articles, author bio. Inside an <article> it relates to that article; directly inside <main> it relates to the whole page.

<footer>

Closing content for a page or section — copyright, links, contact info. Like <header>, a page can have multiple footers (one per article or section).

```
<!-- Full semantic page skeleton -->
<body>
  <header>
    <nav aria-label="Primary">...</nav>
  </header>

  <main>
    <section>
      <h2>Section Title</h2>
      <article>
        <header><h3>Article Title</h3></header>
        <p>Article content...</p>
        <footer>Tags, date, author</footer>
      </article>
    </section>
  </main>

  <aside>Related links / sidebar</aside>

  <footer>Copyright, links, contact</footer>
</body>
```

More Semantic Elements

Element	Purpose
<figure>	Wraps self-contained content (image, diagram, code) referenced from text.
<figcaption>	A visible caption for a <figure> element.

Element	Purpose
<time>	A specific date or time. Use datetime attribute for machine-readable format.
<address>	Contact information for the nearest ancestor article or page body.
<details>	A disclosure accordion — collapsed by default, expanded on click (no JS needed).
<summary>	The visible heading/label for a <details> element.
<dialog>	A modal or popup dialog box. Controlled via showModal() in JavaScript.
<progress>	Shows the completion progress of a task (e.g. file upload bar).
<meter>	Shows a scalar value within a known range (with low/high/optimum thresholds).
<blockquote>	A long quotation from an external source. Use cite attribute for URL.
<abbr>	An abbreviation. Use title attribute for the full expansion on hover.
<mark>	Highlighted text — relevant to the user's current context or search.
	Content that has been deleted or removed.
<ins>	Content that has been inserted or added.
<kbd>	Keyboard input — keys the user should press. e.g. Ctrl+S

Part 5 — Key HTML Attributes

Global Attributes — Work on Any Element

Attribute	Purpose
id	Unique identifier — one per page. Used by CSS (<code>#myId</code>) and JS (<code>getElementById</code>).
class	One or more class names (space-separated). Used by CSS and JS for grouping.
style	Inline CSS. Use sparingly — prefer external CSS stylesheets.
title	Tooltip text shown when user hovers over the element.
hidden	Hides the element — like <code>display:none</code> but with semantic meaning.
tabindex	Controls keyboard tab order. 0 = focusable, -1 = remove from tab order.
data-*	Custom data attributes. Store extra info for JavaScript. e.g. <code>data-user-id="42"</code>
contenteditable	Makes the element's content editable directly in the browser.
lang	Declares the language of the element's content. e.g. <code>lang="sw"</code> for Swahili.

ARIA Accessibility Attributes

ARIA Attribute	What It Does
aria-label	Provides an accessible name when no visible text label exists.
aria-labelledby	Points to the ID of another element that acts as the label.
aria-hidden	Hides element from screen readers (for decorative icons).
aria-expanded	Indicates if a collapsible element is open (true) or closed (false).
aria-current	Marks the current item in a set. e.g. <code>aria-current="page"</code> on active nav link.
aria-live	Announces dynamic content changes. Values: polite or assertive.
role	Overrides the element's default role. Only use when semantic HTML cannot.

Part 6 — Best Practices & Accessibility

HTML Best Practices

- Use lowercase tags — always write `<section>` not `<SECTION>`.
- Close all tags — every non-self-closing tag needs a closing tag.
- Quote all attributes — always use `href="value"` not `href=value`.
- Use semantic elements — `<header>`, `<nav>`, `<main>` instead of `<div class="header">`.
- One `<h1>` per page — exactly one, and it must be the main topic.
- Never skip heading levels — go `h1` → `h2` → `h3`, never `h1` → `h3`.
- Always add alt to images — empty `alt=""` for decorative-only images.
- Label every input — use `<label for="id">` linked to the input's id.
- Validate your HTML — use validator.w3.org to check for errors.
- Separate concerns — HTML for structure, CSS for style, JS for behaviour.
- No inline styles — avoid `style="..."` in HTML, write CSS in stylesheets.
- Add lang to `<html>` — `lang="en"` helps screen readers and search engines.
- Use `loading="lazy"` on images — defers off-screen images, speeds up page load.

Accessibility Checklist

Quick Accessibility Checklist

- ☐ Every image has a meaningful alt attribute (or `alt=""` if decorative)
- ☐ Every form input has an associated `<label>`
- ☐ The page has exactly one `<h1>` and headings are in logical order
- ☐ Semantic elements are used instead of generic `<div>` wherever possible
- ☐ All links have descriptive text (not 'click here' or 'read more')
- ☐ Colour contrast meets WCAG AA standards (4.5:1 for normal text)
- ☐ The page can be fully navigated using only the keyboard
- ☐ `lang` attribute is set on the `<html>` element
- ☐ The page title is descriptive and unique
- ☐ No content relies on colour alone to convey meaning

Write code every day. Read code every day. You will be amazed at how fast you grow.

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